补：对角阵的性质

性质1：若 $\Lambda = diag(\lambda_1, \lambda_2, \cdots, \lambda_m)$，则 $\Lambda^{-1} = diag(\lambda_1^{-1}, \lambda_2^{-1}, \cdots, \lambda_m^{-1})$

性质2：若 $\Lambda = \boldsymbol{diag}(\lambda_1, \lambda_2, \cdots, \lambda_{\boldsymbol{m}})$，则 $\Lambda^{\boldsymbol{n}} = \boldsymbol{diag}(\lambda_1^{\boldsymbol{n}}, \lambda_2^{\boldsymbol{n}}, \cdots, \lambda_{\boldsymbol{m}}^{\boldsymbol{n}})$

即 $\Lambda^{\boldsymbol{n}} = \begin{pmatrix} \lambda_1^{\boldsymbol{n}} & & & \\ & \lambda_2^{\boldsymbol{n}} & & \\ & & \ddots & \\ & & & \lambda_{\boldsymbol{m}}^{\boldsymbol{n}} \end{pmatrix}$

性质3：若 $A = diag(a_1, a_2, \cdots, a_m)$，$B = diag(b_1, b_2, \cdots, b_m)$

则 $A \pm B = diag(a_1 \pm b_1, a_2 \pm b_2, \cdots, a_m \pm b_m)$

$\lambda A = diag(\lambda a_1, \lambda a_2, \cdots, \lambda a_m)$

$AB = diag(a_1 b_1, a_2 b_2, \cdots, a_m b_m)$

补：对角阵的性质

性质4：若 $A=\begin{pmatrix} a_1 & & & \\ & a_2 & & \\ & & \ddots & \\ & & & a_n \end{pmatrix}$，$B=\begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{pmatrix}$

则 $AB=\begin{pmatrix} a_1b_{11} & a_1b_{12} & \cdots & a_1b_{1n} \\ a_2b_{21} & a_2b_{22} & \cdots & a_2b_{2n} \\ \vdots & \vdots & & \vdots \\ a_nb_{n1} & a_nb_{n2} & \cdots & a_nb_{nn} \end{pmatrix}$，$BA=\begin{pmatrix} a_1b_{11} & a_2b_{12} & \cdots & a_nb_{1n} \\ a_1b_{21} & a_2b_{22} & \cdots & a_nb_{2n} \\ \vdots & \vdots & & \vdots \\ a_1b_{n1} & a_2b_{n2} & \cdots & a_nb_{nn} \end{pmatrix}$

- 左乘对角阵，等价于每行乘以对应对角元素
- 右乘对角阵，等价于每列乘以对应对角元素

“左行右列”

补：对角阵的性质

性质5：设 $\varphi(x)$ 是 x 的 m 次多项式，$\varphi(x)=a_m x^m+\cdots+a_1x+a_0 \quad (a_m\neq 0)$

若 $\Lambda=diag(\lambda_1,\lambda_2,\cdots,\lambda_s)$，则

$$\varphi(\Lambda)=a_m\Lambda^m+\cdots+a_1\Lambda+a_0E$$

$$=a_m\begin{pmatrix}\lambda_1^m & & & \\ & \lambda_2^m & & \\ & & \ddots & \\ & & & \lambda_s^m\end{pmatrix}+\cdots+a_1\begin{pmatrix}\lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_s\end{pmatrix}+a_0\begin{pmatrix}1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1\end{pmatrix}$$

$$=\begin{pmatrix}a_m\lambda_1^m+\cdots+a_1\lambda_1^m+a_0 & & & \\ & a_m\lambda_2^m+\cdots+a_1\lambda_2^m+a_0 & & \\ & & \ddots & \\ & & & a_m\lambda_s^m+\cdots+a_1\lambda_s^m+a_0\end{pmatrix}=\begin{pmatrix}\varphi(\lambda_1) & & & \\ & \varphi(\lambda_2) & & \\ & & \ddots & \\ & & & \varphi(\lambda_s)\end{pmatrix}$$

练习题：设三阶矩阵A, B满足关系：

$$A^{-1}BA = 6A + BA, \text{且} A = \begin{pmatrix} 1/2 & & \\ & 1/4 & \\ & & 1/7 \end{pmatrix}, \text{求} B.$$

解：$A^{-1}BA - BA = 6A$

$$\Rightarrow \left(A^{-1} - E\right)B = 6E$$

$$\Rightarrow B = \left(A^{-1} - E\right)^{-1} \cdot 6E = 6\left(A^{-1} - E\right)^{-1}$$

$$= 6\begin{pmatrix} 2-1 & & \\ & 4-1 & \\ & & 7-1 \end{pmatrix}^{-1} = 6\begin{pmatrix} 1 & & \\ & 3 & \\ & & 6 \end{pmatrix}^{-1} = \begin{pmatrix} 6 & & \\ & 2 & \\ & & 1 \end{pmatrix}$$